The diamond conditions for unconfined systems and a minimal pair

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Editorial note. The counted statement is the author's environment at source0.tex lines 650-655 and its proof environment opens at line 657; the pairing is by adjacency and the retained proof range is the whole environment.

Editorial note. The preamble is the author's own package lines, macro block and theorem-environment declarations (source lines 4-65); the author's optioned amsmath line and the fleqn-dependent layout line are not retained.

Editorial note. The source carries no label-number dump and no counter values are imposed, so the printed numbering is the wrapper's default; identity is the author label, or the pinned source position where the author gave none.

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Editorial note. r002 note: r001 failed because the retained author proof cites two bibliography keys of the article; r002 bundles the article's own bibliography file and changes nothing else.

Editorial scope. One original result of the pinned author manuscript, counted once: the two diamond conditions for unconfined systems (author label diamond).

Allowed standard prerequisites (editorial).

- HF-ordering. A linear ordering in which every element has at most one pair of smaller elements forming a block with it.. It is what the cited proposition supplies for an unconfined system.

- Elementary substructure. A$\leq$B means that A is an elementary substructure of B, so that the closure conditions transfer.. It is the relation appearing in the second item.
- Inherited ordering. A subset of an ordered structure inherits the ordering, and the initial segment condition restricts to it.. It is the mechanism of the second item.

Proposition 0.1 1. Free Steiner triple systems are unconfined.

2. Finitely generated unconfined STSs are free.
3. A partial STS is unconfined if and only if it has an HF-ordering.

Lemma 0.2 Let $A, A_1, A_2 \in \mathcal{C}_{\text{un}}$. Then

1. $\emptyset \leq A$
2. If $B \in \mathcal{C}_{\text{un}}$ is such that $A_1 \subseteq B$ and $A_2 \leq B$, then $A_1 \cap A_2 \leq A_1$.

Proof. 1. By Proposition 0.1, since every $A \in \mathcal{C}_{\text{un}}$ has an HF-ordering.

2. Given an HF-ordering of B with initial segment A_2, the elements of A_1 inherit an HF-ordering with initial segment $A_1 \cap A_2$. □

References

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